



SERIES — S

12V**SODIUM-ION BATTERY**

Sodium-Ion NFPP technology engineered for RV, marine, off-grid, and mobility applications — with superior safety and maintenance-free operation.

**30-
100^{Ah}**CAPACITY
RANGE**12^V**

NOMINAL VOLTAGE

3,000+

CYCLE LIFE

40%LIGHTER VS.
LEAD-ACID**-40/+60°C**

OPERATING TEMP

Zero

MAINTENANCE

— ENERGY STORAGE YOU CAN BANK ON

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UL 1973 UN 38.3 IEC 62619 RoHS CE Air-Shippable

PRODUCT OVERVIEW

12V Sodium-Ion Battery Pack



12V 30 / 60 / 100Ah Sodium-Ion Battery Pack

The ultimate 12V battery solution for RV, marine, off-grid, and mobility applications. Available in 30Ah, 60Ah, and 100Ah capacities to match your power needs. Sodium-ion chemistry delivers superior safety, extreme temperature performance, and maintenance-free operation — making it the ideal drop-in replacement for lead-acid batteries.

COMPLETE BATTERY SYSTEM INCLUDES:

- Sodium-ion prismatic cells (4S configuration)
- Rugged weatherproof enclosure
- State-of-charge LED indicator
- Integrated BMS with protective circuitry
- Standard 12V terminals with protective covers
- Over-voltage & short-circuit protection

KEY BENEFITS



Safety-First Design

Sodium-ion chemistry eliminates fire risks associated with lithium-ion batteries — enabling safe use in enclosed spaces.

- Zero thermal runaway risk
- Non-hazardous air shipping
- No toxic gas emissions
- Safe for enclosed compartments



Superior Performance

Outperforms both lithium-ion and lead-acid in demanding conditions — from arctic cold to desert heat.

- 3,000+ cycles at 80% DoD
- -40°C to +60°C operation
- 40% lighter than lead-acid
- Fast charging capability



Maintenance-Free

True drop-in replacement requiring no special equipment or ongoing maintenance — just install and use.

- No watering or equalization
- No sulfation or memory effect
- Works with standard 12V chargers
- Built-in BMS protection

WHERE THE SERIES S PERFORMS





TECHNICAL DATA SHEET

Specifications



3

CAPACITY OPTIONS (30/60/100 AH)

12V

NOMINAL VOLTAGE

3,000+

CYCLE LIFE @ 80% DOD

-40°C

MINIMUM OPERATING TEMP

TECHNICAL SPECIFICATIONS — ALL MODELS

PARAMETER	30AH MODEL	60AH MODEL	100AH MODEL
Model Number	COL-12-30	COL-12-60	COL-12-100
Nominal Voltage	12V	12V	12V
Capacity	30 Ah	60 Ah	100 Ah
Energy	360 Wh	720 Wh	1,200 Wh
Cell Configuration	4S3P	4S6P	4S10P
Dimensions (L×W×H mm)	196×128×220	264×167×220	328×172×220
Weight	6.5 kg / 14.3 lb	9 kg / 19.8 lb	13 kg / 28.7 lb
Cycle Life	≥3,000 @ 1C 80% DoD / ≥5,000 @ 0.2C		
Charge Temperature	-10°C to +45°C		
Discharge Temperature	-40°C to +60°C		
Voltage Range	8V – 15.8V		
Max Discharge Rate	1.0C continuous		
Typical Runtime	12–24 hrs	1–2 days	2–4 days
Best For	Trolling motors, portable power	RV accessories, solar systems	House batteries, full RV setups



30Ah Model
COL-12-30 · 360 Wh



60Ah Model
COL-12-60 · 720 Wh



100Ah Model
COL-12-100 · 1,200 Wh

Series S — 12V Sodium-Ion Battery Pack | Rev. 1.0

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TECHNOLOGY COMPARISON

Sodium-Ion NFPP vs. Lead-Acid



A Clear Advantage Across Every Performance Dimension

Coulomb Technology's Sodium-Ion NFPP chemistry delivers superior cycle life, wider operating temperature range, zero maintenance, and a fundamentally safer chemistry — making it the ideal replacement for lead-acid in RV, marine, off-grid, and mobility applications.

FEATURE	SODIUM-ION NFPP	LEAD-ACID
Cycle Life	3,000+ cycles @ 80% DoD	300-500 cycles @ 50% DoD
Weight (100Ah)	13 kg (28.7 lb)	~32 kg (70.5 lb)
Usable Capacity	80-100% DoD	50% DoD (to preserve life)
Discharge Temp Range	-40°C to +60°C	-20°C to +50°C
Charge Temp Range	-10°C to +45°C	0°C to +40°C
Maintenance	Zero maintenance	Regular watering & equalization
Self-Discharge	<3% per month	5-15% per month
Charging Efficiency	~95%	~75%
Toxic Gas Emissions	None	Hydrogen gas during charging
Memory Effect	None	Sulfation if undercharged
Lifespan	10+ years	3-5 years
Environmental Impact	Non-toxic, recyclable	Lead & acid — hazardous

3 WAYS TO CHARGE



12V Battery Charger

Use any standard 12V sodium-ion or LiFePO4-compatible charger. Fully charges in 2-10 hours depending on capacity.



Solar Panels (≥100W)

Pairs seamlessly with any 12V solar charge controller. Recommended minimum 100W panel for efficient charging.



Generator + DC Charger

Connect a 12V DC charger to any AC generator for reliable charging when solar or shore power is unavailable.

10×

Longer cycle life
vs. lead-acid

40%

Lighter weight
per 100Ah

Zero

Maintenance
required

95%

Round-trip
efficiency

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APPLICATION SCENARIOS

Where the Series S Performs



Built for Life Off the Grid

The Series S 12V battery delivers reliable, maintenance-free power wherever you need it — from weekend camping trips to full-time off-grid living. Its extreme temperature tolerance and superior cycle life make it the ideal drop-in replacement for lead-acid in any 12V application.



RV & MOTORHOME

RV & Camping Power

Power lights, fans, water pumps, and 12V appliances for extended off-grid camping trips. The lightweight design reduces vehicle weight compared to lead-acid batteries.

- Reliable power for extended off-grid trips
- Lightweight — reduces vehicle weight
- Works with existing 12V systems



MARINE & BOATING

Marine & Boat Power

Power trolling motors, fish finders, and navigation equipment in harsh marine environments. Vibration-resistant and corrosion-free with no toxic gas emissions.

- Safe for enclosed boat battery compartments
- Vibration-resistant construction
- No toxic gas emissions



OFF-GRID & SOLAR
Off-Grid Solar Storage



MOBILITY & ACCESSIBILITY
Mobility & Scooters